

Project Workflow - Human Development Index (HDI) Prediction Using Machine Learning

Step 1: Dataset Collection

Collect the HDI dataset from trusted sources such as UNDP, World Bank, or Kaggle.

Step 2: Data Preprocessing

Clean the dataset, handle missing values, remove duplicates, normalize data, and select relevant features.

Step 3: Exploratory Data Analysis

Analyze the dataset using statistics and visualizations to identify patterns and relationships.

Step 4: Dataset Splitting

Split the dataset into training and testing sets, typically using an 80:20 ratio.

Step 5: Model Training

Train machine learning regression models such as Linear Regression, Decision Tree, Random Forest, or Gradient Boosting.

Step 6: Model Evaluation

Evaluate the models using MAE, MSE, RMSE, and R^2 Score, then select the best-performing model.

Step 7: Model Saving

Save the trained model as a .pkl file using Joblib or Pickle.

Step 8: Web Application Development

Develop a Flask-based web application with HTML/CSS for user interaction.

Step 9: Prediction

Accept user input values, process them through the trained model, and predict the HDI value.

Step 10: Results and Visualization

Display the predicted HDI along with charts, reports, and visual insights.

Workflow Summary

Dataset Collection → Data Preprocessing → Exploratory Data Analysis → Dataset Splitting → Model Training → Model Evaluation → Model Saving → Flask Web Application → User Input → HDI Prediction → Results & Visualization